

Writing Small Numbers in Scientific Notation

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.9 (SE pp. 146–151) and completes MA.8.NSO.1.4. The book's sequence — negative powers of ten and decimal places, the definition for numbers less than 1, then 'how many times larger or smaller' — is kept as Notes I–III. Its oxygen density, human hair, Mycoplasma, 0.0000988 and $7.2 \times 10^{-4} \div 1.8 \times 10^{-7}$ items are kept.

Three book items were NOT reused, per the audit. Homework 4b's key says 10^{-8} is 3 times smaller than 10^{-5} (it is 1,000 times). Homework 4c asks how many times smaller 7.4×10^{-8} is than 0.000039 and keys 3,000; the true ratio is about 527, and the item's scaffold pulls in two directions — bank 8 keeps the same two numbers and asks only which is greater, which is sound. Homework 5's matching pairs '112 ten millionths' with $1.12 \cdot 10^{-7}$, but 112 ten-millionths is $0.0000112 = 1.12 \times 10^{-5}$, the same value as another row; Additional 10 uses '112 billionths', which is 1.12×10^{-7} .

Also not reused: Additional Practice 3's first row, which prints as $(10^4) - 4 \times 10^{25}$ and keys -4×10^{25} exactly (the 10^4 was dropped); the item is malformed on the page.

Added from the state guide: comparisons that cross from very large to very small (Notes III line 2, bank 13, Additional 13), E notation on the small side (whiteboard 8), and the fishing-line task as whiteboard 9, written.

Every scientific-notation coefficient in this lesson is in $[1, 10)$ except in the items that are ABOUT the constraint (whiteboards 4–5; bank 11–12 and their Additional twins). Those are tagged in the spec and were checked by hand.

Standard and Targets

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Benchmark Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Boundary Both directions of comparison ('times larger' and 'times smaller') are in the benchmark; both are in the round and the bank. Operations in scientific notation are MA.8.NSO.1.5, Unit 4.

Learning target I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

Essential question How does a negative power of 10 let us write a very small number, and compare it to another?

Building on 3.04 — the negative exponent law, $10^{-n} = 1/10^n$; 3.08 — scientific notation for large numbers; Grade 5 — decimal place value to the thousandths.

Working toward MA.8.NSO.1.5 (Unit 4) — adding, subtracting, multiplying and dividing in scientific notation; MA.8.NSO.1.6 — significant digits.

Where this sits in the course Second of two MA.8.NSO.1.4 lessons; the benchmark is complete after today, and the unit's content is complete — the review and the unit assessment follow. The Grade 8 FAST reports this under Number Sense and Operations.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; entering 4.2×10^{-6} and reading the display back as 4.2E-6 is worth thirty seconds at whiteboard 8.

Where the lesson lives Notes I — 'the exponent tells you which decimal place the 1 is in, not how many zeros'. Whiteboard 5 is the diagnostic (option B is the zero-counter); whiteboard 9 asks for a comparison in writing.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

negative power of 10 — $10^{-n} = \frac{1}{10^n}$: a 1 in the n th decimal place.
 $10^{-3} = \frac{1}{1,000} = 0.001$

scientific notation — $a \times 10^n$ with $1 \leq a < 10$ and n an integer — for a number less than 1, n is negative: $0.00025 = 2.5 \times 10^{-4}$

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Write 4,300,000 in scientific notation. **4.3×10^6**

retrieves: yesterday — 3.08 — the 4 is in the millions place

2. Write 10^{-3} as a fraction. **$\frac{1}{1,000}$**

retrieves: last week — 3.04 — the negative exponent law with base 10

3. $\sqrt{95}$ is between which two whole numbers? **9 and 10**

retrieves: last unit — Unit 2 — locating a square root (2.05): $81 < 95 < 100$

4. Write 0.007 as a fraction. **$\frac{7}{1,000}$**

retrieves: prior grade — the prerequisite this lesson needs — Grade 5 — decimal place value (MA.5.NSO.1): the 7 is in the thousandths place

Slide-by-Slide Teaching Notes

Slide 1. Writing Small Numbers in Scientific Notation 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q2 and Q4 are the two halves of today: 10^{-3} is a thousandth, and 0.007 is 7 thousandths, so 0.007 is 7×10^{-3} . Say that at the reveal. Q1 is yesterday's skill and today's is the mirror image.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. 3 min

- The 3.04 law with base 10: 10^{-n} is 1 over 10^n , and as a decimal it is a 1 in the n th decimal place. 10^{-3} is one thousandth, 0.001, and the 1 is in the thousandths place. Students fill the last two cells. The count to watch: 0.001 has two zeros after the point and the 1 in the THIRD place — the exponent counts places, not zeros.
- **OFF THE SLIDE, YOURS TO SAY:** 'The exponent tells you which decimal place the 1 is in. Not how many zeros.'

Slide 5. Notes 4 min

- Same definition as yesterday; the exponent is negative because the number is less than 1. A red blood cell is 0.000007 m across: the 7 is in the millionths place, so 7×10^{-6} . Then 0.0000672: the first nonzero digit, 6, is in the hundred-thousandths place, so 6.72×10^{-5} — and it has only four zeros after the point. Reading back: 3.4×10^{-9} puts the 3 in the billionths place, eight zeros then 34. The rule of thumb: the point moves right n places, the exponent is $-n$.
- **OFF THE SLIDE, YOURS TO SAY:** 'Find the first nonzero digit. Its place is the exponent. Everything else is the coefficient.'

Slide 6. Notes 4 min

- Same routine as yesterday; the only new thing is that dividing powers of 10 with negative exponents is integer subtraction: $10^{-4} \div 10^{-7} = 10^3$. Line 1 is the book's collaboration item, and it is sound. Line 2 crosses from a large number to a small one — the state guide's item shape — and the exponent difference $8 - (-4) = 12$ does all the work. 'Times smaller' is the same division read the other way: 1.8×10^{-7} is 4,000 times smaller than 7.2×10^{-4} .
- **OFF THE SLIDE, YOURS TO SAY:** 'The larger exponent wins. -4 is larger than -7 , so 10^{-4} is the bigger power.'

Slide 7. Example 1 — A quantity in words, to scientific notation. 1 min

- SPOKEN ONLY. The book's oxygen density. Two steps: words to a decimal, decimal to scientific notation. Ask what 'millionths' means as a place before anyone writes.

Slide 8. Example 1 — Worked. 2 min

- 1,332 millionths is $1,332 \div 1,000,000 = 0.001332$. The first nonzero digit, 1, is in the thousandths place: 1.332×10^{-3} . A board with 1.332×10^{-6} used the word 'millionths' as the exponent without converting; a board with 1.332×10^{-2} counted the zeros.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- The book's human hair. 67 hundred-thousandths = $0.00067 = 6.7 \times 10^{-4}$. Catches 6.7×10^{-5} ('hundred-thousandths' used as the exponent) and 6.7×10^{-3} (zeros counted).

Slide 10. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — How many times larger? 1 min

- SPOKEN ONLY. The book's collaboration item. Ask which number is larger before anyone divides — the exponents decide, and -4 beats -7 .

Slide 12. Example 2 — Worked. 2 min

- $7.2 \div 1.8 = 4$ and $10^{-4} \div 10^{-7} = 10^3$ (-4 minus -7 is 3). So 4,000 times. A board with 4 stopped at the coefficients; a board with 4×10^{-3} subtracted the exponents the wrong way ($-7 - (-4)$); a board with 40 wrote 10 for 10^3 . Confirm: $0.00072 \div 0.00000018 = 4,000$.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- The book's practice item. $5.2 \div 2.6 = 2$ and $10^{-3} \div 10^{-8} = 10^5$: 200,000. A board with 2×10^{-5} subtracted the wrong way.

Slide 14. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 *19 min for the round*

- The law with base 10. A board with 0.00001 counted four zeros and then wrote the 1; a board with $-10,000$ read the exponent's sign as the number's sign.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- One digit. 3×10^3 has the sign wrong (that is 3,000); 3×10^{-2} counted the zeros.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- The book's Mycoplasma bacterium, 2.5×10^{-4} mm. A board with 0.000025 moved five places; a board with 25,000 moved the wrong way.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- The book's ticket-out-the-door value. Four zeros after the point but the 9 is in the fifth place: 10^{-5} . A board with 9.88×10^{-4} counted zeros; a board with 988×10^{-7} is equal in value but broke the coefficient rule.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. B is the zero-counter, the most common wrong board today; C is the coefficient rule; D is the sign.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic for the comparison. D is the integer-subtraction slip — the one that says the powers were handled, just backwards.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Powers first. 7.4 is the bigger coefficient, but 10^{-8} is a much smaller power. A board with 7.4×10^{-8} compared the coefficients or thought 8 beats 5.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- The state guide's E-notation shape, small side. A board with $-4,200,000$ read the minus as the number's sign; a board with 0.00000042 moved seven places.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. The state guide's task for this benchmark. Full credit: the thicker line, the factor 3, and a value strictly between 0.02 and 0.06 written in either form. A student who answers '3 times thinner' for the first line has the comparison right and the direction reversed — mark it right only if the sentence says which line is which. A student who gives 4×10^{-1} has the coefficient right and the power wrong (that is 0.4 in, thicker than both).

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	10^{-4}	<i>0.0001</i>	0.00001 — four zeros written, then the 1 (five places) [8.NSO.1.4: the exponent counts places]; $-10,000$ — negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception]
2	Write 0.003 in scientific notation.	<i>3×10^{-3}</i>	3×10^3 — the sign was dropped: that is 3,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]; 3×10^{-2} — counted the zeros instead of the place
3	Write 2.5×10^{-4} in standard form.	<i>0.00025</i>	0.000025 — point moved five places [8.NSO.1.4: 10^{-4} moves the point four places]; 25,000 — point moved right instead of left [negative exponent read as a large number]

#	Question	Answer	What a wrong board says
4	Write 0.0000988 in scientific notation.	9.88×10^{-5}	9.88×10^{-4} — counted the zeros [8.NSO.1.4: exponent = places moved]; 988×10^{-7} — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]
5	Which is 0.00081 written in scientific notation?	$A - 8.1 \times 10^{-4}$	(B) counted the three zeros instead of the four places [8.NSO.1.4: exponent = places moved]; (C) equal in value, but 81 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]; (D) the sign was dropped: 8.1×10^4 is 81,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]
6	How many times larger is 6×10^{-3} than 2×10^{-5} ?	$A - 300$	(B) coefficients only — the powers were ignored [8.NSO.1.4: both factors are compared]; (C) 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: exponent read as the multiplier]; (D) the exponents were subtracted the wrong way, $-5 - (-3) = -2$ [8.NSO.1.3: quotient law, top minus bottom]
7	Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?	3.9×10^{-5}	7.4×10^{-8} — coefficients compared first, or -8 read as larger than -5 [B1G-M 8.NSO.1.4: order of magnitude first; integer ordering]
8	A computer displays a number as $4.2\text{E}-6$. Write the number in standard form.	0.0000042	$-4,200,000$ — the exponent's sign read as the number's sign [B1G-M 8.NSO.1.4 misconception: E notation]; 0.00000042 — point moved seven places
9	Fishing line A is 2×10^{-2} inch thick. Fishing line B is 6×10^{-2} inch thick. Which is thicker, and how many times thicker? Give a possible thickness between the two.	B; 3 times; e.g. 4×10^{-2} in.	' 2×10^{-2} is thicker' — 10^{-2} read as making a smaller coefficient bigger; ' 4×10^{-1} in between' — power ignored in the in-between value [8.NSO.1.4: both factors matter]

Question Bank — variation and answers

Variation. Questions 1–4 write in scientific notation with one, two, three and three significant digits and exponents from -2 to -11 . 5–6 read notation back. 7 is powers of ten alone: a fraction, two laws, one law. 8 pairs a conversion each way with a comparison by inspection. 9 is the scaffolded comparison with negative exponents in the powers. 10 is the book's word-form values with a sound third row. 11 is the diagnostic multiple choice; 12 separates value from form. 13 crosses from large to small and asks the comparison both ways. The Additional sheet mirrors each position.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

1. 5×10^{-4}
2. 4.9×10^{-9}
3. 3.16×10^{-2}
4. 5.29×10^{-11} **m**
5. **0.000009**
6. **0.00000018**
7. (a) $\frac{1}{100,000,000,000}$ (b) $10^{-24} \cdot 10^{22} = 10^{-2} = 0.01$ (c) $10^3 = 1,000$
8. (a) **0.000039** (b) 7.4×10^{-8} (c) 3.9×10^{-5} — *the exponent -5 is greater than -8, so its power of 10 is larger; 0.000039 > 0.000000074.*
9. (a) **3** (b) $10^4 = 10,000$ (c) $3 \times 10^4 = 30,000$ *times larger*
10. (a) $0.0988 = 9.88 \times 10^{-2}$ (b) 9.88×10^{-3} (c) $0.000988 = 9.88 \times 10^{-4}$
11. **A A**
12. **A, B, C, D A, B, C and D**
13. (a) $10^{12} = 1,000,000,000,000$ *times* (b) 10^{12} *times slower — the same comparison read the other way.*

Question Bank — Additional

1. 8×10^{-3}
2. 2.6×10^{-7}
3. 4.75×10^{-2}
4. 9.1×10^{-11} **s**
5. **0.00007**
6. **0.000000026**
7. (a) $\frac{1}{10,000,000,000,000,000}$ (b) $(10^5)^3 = 10^{15} = 1,000,000,000,000,000$ (c) $10^{-3} = 0.001$
8. (a) **0.00072** (b) 1.8×10^{-7} (c) 7.2×10^{-4} — *-4 is greater than -7, so its power of 10 is larger; 0.00072 > 0.00000018.*
9. (a) **4** (b) $10^4 = 10,000$ (c) $4 \times 10^4 = 40,000$ *times larger*
10. (a) $0.0112 = 1.12 \times 10^{-2}$ (b) 1.12×10^{-3} (c) $0.000000112 = 1.12 \times 10^{-7}$
11. **A A**
12. **A, B, C, D A, B, C and D**
13. (a) 2.5×10^{11} *times* (b) 2.5×10^{11} *times smaller — the same comparison read the other way.*

What the audit against Math Nation found

- Every item re-derived by sympy at build time (mathcheck: 0 findings); every scientific-notation coefficient scanned (capcheck: 0 findings outside tagged items).

- Math Nation 3.9 student pages verified: $10^{-12} = 1/10^{12}$, $10^{-7} = 1/10^7$, 7×10^{-6} , 6.72×10^{-5} , 0.0000000034, 1.332×10^{-3} , 0.00025, $7.2 \times 10^{-4} \div 1.8 \times 10^{-7} = 4,000$, the five ticket-out-the-door matches.
- Homework L9 #4b key (3) is wrong — 1,000 (audit D5). #4c (3,000) is wrong and the item is unsound — true ratio ≈ 527 (audit D6). #5 has two left-hand values that are equal (audit D7). None reused.
- Additional Practice L9 #3 row 1 malformed on the page and keyed wrong (audit D3). Not reused. Rows 2–3 ($(1/10^{-5})^3 = 10^{15}$; $10^{-9} \cdot 10^6 = 10^{-3}$ — the key prints $10^{-5} \cdot 10^6 = 10$, a different item) — our Additional 7b–c use the sound versions.
- B1G–M items verified: $0.0000000000529 = 5.29 \times 10^{-11}$; fishing line $6 \times 10^{-2} \div 2 \times 10^{-2} = 3$; $3 \times 10^8 \div 3 \times 10^{-4} = 10^{12}$.

What changed from the book

- 3.9.1 googol warm-up (a googol squared has 200 zeros) → spaced retrieval; the googol is a fine aside if a minute is spare.
- 3.9.2 partner exploration (powers of ten) → Notes I from the front.
- 3.9.3 Bring It Together (laws with powers of ten) → bank 7.
- 3.9.4–3.9.5 Guided Instruction and Guided Practice → Notes II and Example 1.
- 3.9.6 partner collaboration (comparing) → Notes III, Example 2 and whiteboards 6–7.
- 3.9.7 ticket-out-the-door matching → bank 10, as write-the-notation items.